Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-6 | Summar-2025

Course Code	:	2301CS632	**Date**	:	15-04-2025
Course Name	:	Machine Learning	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Differentiate supervised, unsupervised and reinforcement learning.	**4**
	(B)	Differentiate Model-Based and Instance-Based Learning.	**3**
		OR	
		Explain algorithm, model and features with respect to machine learning.	
	(C)	Explain Machine Learning project life cycle.	**7**
		OR	
		Explain main challenges of machine learning.	
Q.2	**(A)**	Differentiate L1 and L2 regularization.	**4**
	(B)	Explain kernel trick in SVM.	**3**
		OR	
		Explain R square with an example.	
	(C)	Write a python code to Implement a Linear Regression model to predict house prices based on the number of bedrooms, plot the regression line, and predict the price for a house with 5 bedrooms.	**7**
		OR	
		Write a python code to implement logistic regression to classify iris flower.	
Q.3	**(A)**	Explain K- Nearest neighbours algorithm with an example.	**4**
	(B)	Differentiate bagging and boosting.	**3**
		OR	
		Explain confusion matrix with an example.	
	(C)	Explain decision tree algorithm with an example.	**7**
		OR	
		Explain Adaboost algorithm with an example.	

Q.4	**(A)**	Write a Python program to implement agglomerative clustering with make_blobs for data generation and display dendrogram.	**4**
	(B)	Explain K-mean clustering with an example.	**3**
		OR	
		Explain filter methods for feature selection.	
	(C)	What is PCA? Explain steps in PCA.	**7**
		OR	
		Explain DBSCAN algorithm with an example.	
Q.5	**(A)**	Explain basic components of perceptron with example.	**4**
	(B)	Explain Multi-layer feed-forward networks.	**3**
		OR	
		Explain delta rule.	
	(C)	Explain all activation functions with example.	**7**
		OR	
		Explain hyperparameters with example.	

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